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Aircraft comprising a dihydrogen transport line, a channel for embedding said transport line and a system for evacuating air and water present in said embedded channel

Abstract

An aircraft with a dihydrogen transport line, an embedded channel having a gutter in which the transport line is fixed and having a first end by which the transport line enters into the gutter and a second end by which the transport line leaves the gutter, and an evacuation system fluidically connected between the second end of the gutter and the outside of the aircraft for the evacuation of air and of water.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) This application claims the benefit of French Patent Application Number 2303001 filed on Mar. 29, 2023, the entire disclosure of which is incorporated herein by way of reference.

FIELD OF THE INVENTION

(2) The present invention relates to the field of aircraft and, in particular, aircraft for which the energy source is liquid or gaseous dihydrogen, whether for supplying a fuel cell or directly supplying the combustion chamber of an engine. The present invention thus relates to an aircraft comprising a dihydrogen transport line, an embedded channel which allows the passage of the transport line and a system for evacuating water and air present in the embedded channel.

BACKGROUND OF THE INVENTION

- (3) It is known practice to use dihydrogen as energy source in an aircraft. The dihydrogen is stored in a tank and a transport line ensures the transportation of the dihydrogen from the tank to the consumer device, such as, for example, a fuel cell or the combustion chamber of an engine.
- (4) To secure the transportation of the dihydrogen, an embedded channel in which the transport line is housed and fixed can be used for each transport line.
- (5) In this type of installation, water may have stagnated in the embedded channel and it may be that dihydrogen escapes from a transport line.
- (6) It is therefore desirable to provide an installation which ensures the evacuation of the water and of the air or of the dihydrogen present in the embedded channel to the outside.

SUMMARY OF THE INVENTION

- (7) One object of the present invention is to propose an aircraft comprising a dihydrogen transport line, an embedded channel which allows the passage of the transport line and a system for evacuating air and water and dihydrogen in the event of a leak.
- (8) To this end, an aircraft is proposed comprising: at least one transport line in which dihydrogen flows, between, upstream a tank and downstream a consumer device, for each transport line, an embedded channel comprising a gutter in which the transport line is fixed and having a first end by which the transport line coming from upstream enters into the gutter and a second end by which the transport line going downstream leaves the gutter, and for each second end, an evacuation system fluidically connected between the second end of the gutter and the outside of the aircraft and intended to ensure the evacuation of air and water.
- (9) With such an arrangement, the water and the air present in the transport line are sucked over the entire length of said transport line. Indeed, with such an arrangement, the circulation of air in the transport line is forced, by virtue of a pressurization at the first end of the transport line so as to force the air to circulate in the transport line, and of the resulting suction of the air in the transport line to the second end of the transport line (creating a depression). This makes it possible to maximize a dilution of dihydrogen in the air in the event of a dihydrogen leak.
- (10) Advantageously, the evacuation system comprises a first line with a proximal end, a distal end and a bend between said proximal end and said distal end, the bend constitutes a low point for the first line, and the first line is fluidically connected to the second end of the gutter by said proximal end and emerges outside at an outer wall of the aircraft by said distal end.
- (11) Advantageously, the aircraft comprises a wing and the distal end of the first line emerges on the upper surface of the wing.
- (12) Advantageously, the evacuation system comprises a cap which extends outside of the aircraft around the distal end by having a closed face oriented towards the front of the aircraft and an open face oriented towards the rear of the aircraft.
- (13) Advantageously, the evacuation system comprises a second line with a proximal end and a distal end, the second line is fluidically connected to the bend by said proximal end and emerges outside of the aircraft at an outer wall of the aircraft by said distal end, and, between the proximal end of the second line and the distal end of the second line, the second line has a downward slope.
- (14) Advantageously, the evacuation system comprises a heating element disposed close to said second line.
- (15) Advantageously, the distal end of the second line is oriented towards the rear of the aircraft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The features of the invention mentioned above, and others, will become more clearly apparent on reading the following description of an exemplary embodiment, said description being given in

relation to the attached drawings, in which:

- (2) FIG. 1 is a top view of an aircraft according to the invention,
- (3) FIG. 2 is a cross-sectional view of a wing of the aircraft of FIG. 1 along the line II-II,
- (4) FIG. 3 is a perspective view of an embedded channel according to the invention,
- (5) FIG. 4 is a perspective view of an example of embedded channels and of an aeration system that can be implemented in the invention,
- (6) FIG. 5 is a perspective view of an air and water evacuation system according to the invention,
- (7) FIG. 6 is a cross-sectional view along the line VI-VI of the evacuation system of FIG. 5, and
- (8) FIG. 7 is a cross-sectional view along the line VII-VII of the evacuation system of FIG. 5.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (9) FIG. 1 shows an aircraft **100** which has a fuselage **102** on either side of which is fixed a wing **104**. Under each wing **104** there is fixed at least one propulsion system **106**.
- (10) By convention, X denotes the longitudinal direction of the aircraft **100**, Y the transverse direction of the aircraft **100** which is horizontal when the aircraft is on the ground, and Z the vertical direction or vertical height when the aircraft is on the ground, these three directions X, Y and Z being mutually orthogonal.
- (11) Also, the terms “front” and “rear” should be considered with respect to a direction of advance of the aircraft **100** when the propulsion systems **106** are operating, this direction being represented schematically by the arrow **107**.
- (12) In the embodiment of the invention presented here, each propulsion system **106** comprises an electric engine, a propeller **108** mounted on the engine shaft of said electric engine and a fuel cell which supplies the motor with electricity.
- (13) The fuel cell is supplied with oxygen and dihydrogen in order to produce electricity.
- (14) In another embodiment, the propulsion system **106** can take the form of a jet engine whose fuel is burnt in the combustion chamber and is dihydrogen.
- (15) The aircraft **100** also comprises a dihydrogen tank **110** which is disposed here in a rear part of the fuselage, but which could be disposed in another part of the aircraft **100**. The dihydrogen can be liquid or gaseous.
- (16) To transport the dihydrogen, the aircraft **100** comprises at least one transport line **112a-b** in which the dihydrogen flows, between, upstream the tank **110** and downstream the consumer device **106** which consumes the dihydrogen and which can here be the fuel cell or the jet engine.
- (17) For simplicity of implementation, the transport lines **112a-b** are preferentially disposed in the upper part of the aircraft **100** and thus extend along the fuselage **102** and the wings **104** in the upper part thereof.
- (18) In the embodiment of the invention presented in FIG. 1, there is a transport line **112a** which extends and ensures the transport of the dihydrogen between the tank **110** and a consumer device **106** on the port side and there is a transport line **112b** which extends and ensures the transport of the dihydrogen between the tank **110** and a consumer device **106** on the starboard side.
- (19) FIG. 2 shows a cross-section of the port-side wing **104** with the transport line **112a**, but the invention applies in the same way to the starboard side. In the embodiment represented in FIG. 2, the aircraft **100** comprises, for each transport line **112a**, an embedded channel **200** comprising a gutter **202** in which the transport line **112a** is housed and fixed.
- (20) FIG. 3 shows the embedded channel **200** with the gutter **202** which, in the embodiment of the invention presented in FIGS. 2 and 3, comprises a bottom **203**, an opening opposite the bottom **203**, and a cover **204** which covers the opening of the gutter **202** by at least partly blocking it. The transport line **112a-b** is thus housed between the bottom **203** and the opening and, here, the cover **204**. The transport line **112a-b** is fixed to the gutter **202** by any appropriate means such as, for example, collars.
- (21) In the embodiment of the invention presented in FIGS. 2 and 3, the gutter **202** has a trapezoidal section, but sections of different forms are possible.

- (22) The cover **204** here constitutes an outer wall of the aircraft **100**, that is to say that it is in direct contact with the outside air surrounding the aircraft **100**. The cover **204** is fixed to the outer wall of the aircraft **100** by any appropriate fixing means, such as, for example here, bolts **210**.
- (23) In the embodiment of the invention presented here, the cover **204** is pierced with ventilation windows **206** which allow the passage of the dihydrogen, particularly in the case of a leak from the transport line **112a-b**, to the outside, limiting equally the concentration of dihydrogen in the embedded channel **200** and the risks linked to the presence of dihydrogen. Preferentially, the cover **204** is in a higher position with respect to the bottom **203**, that is to say on top.
- (24) To limit the impact of the ventilation windows **206** on the drag of the aircraft **100** in flight, each ventilation window **206** is closed by a plug **208** produced in a material that breathes for dihydrogen, that is to say that the plug **208** is impermeable to the outside water that may not enter into the embedded channel **200a-b**, and permeable, among other things, for dihydrogen which is present in the embedded channel **200a-b** and can be evacuated to the outside. The plug **208** is produced for example in polypropylene or polyethylene.
- (25) In the embodiment of the invention presented here, in order to prevent, in the case of fire, the fire from remaining contained in the embedded channel **200** because of the presence of the plugs **208**, each plug **208** is composed of a material that can break under the effect of heat, such as for example polypropylene or polyethylene. By breaking under the effect of heat, the plug **208** constitutes a fusible element which makes it possible to open the ventilation window **206** in case of need to blow out the flames. In particular, the material for the plug **208** is chosen so as to break when the temperature inside the gutter **202** reaches a value lower than the maximum acceptable temperature for the gutter **202** and the transport line **112a-b** installed in the gutter **202**, that is to say the temperature above which the integrity of the gutter **202** and of the transport line **112a-b** is no longer guaranteed.
- (26) In case of overpressure, it is also possible to provide for the cover **204** to break under the effect of the overpressure, thus limiting the risks of breaking of the gutter **202**. To this end, the tear strength of the cover **204** is lower than the tear strength of the gutter **202**, for example through the production of thinner zones. It is also possible to provide for the breaking to be done by means of the cover **204** fixing means, for example by using fusible bolts which break above a certain pressure.
- (27) To aerate each embedded channel **200a-b**, the aircraft **100** comprises an aeration system **600** which is disposed here in the upper part of the aircraft **100** at the join of the wings **104** and which allows the introduction of outside air and allows this air to be sent into each embedded channel **200**.
- (28) To prevent the attachments which fix the transport line **112a-b** in the embedded channel **200** from disrupting the flow of the air coming from the aeration system **600**, the gutter **202** here comprises an intermediate floor **520** between the bottom **203** and the opening. The intermediate floor **520** is at a distance from the bottom **203** and separates the gutter **202** into an upper gutter **521** in which the transport line **112a-b** is fixed and a lower gutter **523** in which the air coming from the aeration system **600** circulates. The upper gutter **521** extends between the intermediate floor **520** and the cover **204**, the lower gutter **523** extends between the bottom **203** and the intermediate floor **520**.
- (29) To allow the air to pass from the lower gutter **523** to the upper gutter **521**, the intermediate floor **520** includes openings and it is, here, pierced with holes **522** which make it possible to ensure a ventilation of the upper gutter **521**.
- (30) FIG. 4 shows an example of the aeration system **600** with the gutters **202** but without the transport lines **112a-b** which are housed in said gutters **202**.
- (31) The aeration system **600** comprises an air vent **610** arranged to capture air outside the aircraft **100** and at least one intake line **614** fluidically connected to the air vent **610**.
- (32) In the embodiment of the invention presented in FIG. 4, there is a plurality of intake lines **614**

grouped together in an intake network **612**, and each intake line **614** channels the air taken from the outside at the air vent **610** to one or more of the lower gutters **523** of the gutters **202**. In the embodiment of the invention presented here, the air vent **610** can alternately take a closed position and an open position by virtue of a hatch **620** which is mounted to be movable between a closed position in which it isolates each window **604** from the outside air and an open position in which it is retracted to freely allow the passage of the outside air to each window **604**. The displacement of the hatch **620** is ensured by any displacement system known to the person skilled in the art such as, for example, a motor or a hydraulic cylinder.

(33) The air vent **610** comprises an air inlet **602** at which each intake line **614** emerges through a window **604** that is open towards the front of the aircraft **100** in order to capture the outside air when the aircraft **100** is advancing or is stopped.

(34) At an intersection between an intake line **614** and one or more of the lower gutters **523**, to ensure the distribution of the air to the or each lower gutter **523**, the aircraft **100** here comprises a junction box **601** which ensures the fluidic connection between the intake line **614** and the or each lower gutter **523**.

(35) Thus, a flow of air entering into the junction box **601** will be diffused at the or each lower gutter **523**.

(36) In the embodiment of the invention presented in FIG. 4, to assist in the flow of the air in the intake lines **614**, particularly when the aircraft **100** is on the ground, fans **650** are installed in one or more of the intake lines **614**.

(37) Each embedded channel **200** thus comprises a first end **630** by which the transport line **112a-b** coming from upstream enters into the gutter **202** and a second end **632** by which the transport line **112a-b** going downstream leaves the gutter **202**. The first end **630** is therefore closest to the tank **110** and the second end **632** is closest to the consumer device **106**.

(38) For each second end **632**, the aircraft **100** comprises an evacuation system **150** fluidically connected between the second end of the gutter **202**, and more particularly here the lower gutter **523** of said gutter **202**, and the outside of the aircraft **100**. The evacuation system **150** which is represented schematically in FIG. 1, is intended to ensure the evacuation of the air and the water which are present in the gutter **202**, and more particularly here in the lower gutter **523** of said gutter **202** and more generally the gases and the liquids present in the gutter **202**. The evacuation system **150** is intended to evacuate only a part of the air present in the gutter **202**. This evacuation system **150** is intended to maximize the flow of the air between the first and second ends **630**, **632** of the embedded channel **200**, in order to maximally dilute any possible leaks of dihydrogen.

(39) FIG. 5 shows an exemplary embodiment of the evacuation system **150** and FIGS. 6 and 7 show cross-sections of said evacuation system **150**. The evacuation system **150** is, here, disposed in the wing **104** in the vicinity of the consumer device **106**, here the propulsion system **106** which is supplied by the transport line **112a**.

(40) The evacuation system **150** comprises a first line **152** which has a proximal end **152a**, a distal end **152b** and a bend **154** between the proximal end **152a** and the distal end **152b**, in which the bend **154** constitutes a low point for the first line **152**, that is to say that, between the proximal end **152a** and the bend **154**, on the one hand, and between the distal end **152b** and the bend **154**, on the other hand, the first line **152** has downward slopes.

(41) The first line **152** is fluidically connected to the second end **632** of the gutter **202**, here of the lower gutter **523** of said gutter **202**, by its proximal end **152a** and emerges outside the aircraft **100** at an outer wall of the aircraft **100**, here the wing **104**, by its distal end **152b**. An outer wall of the aircraft **100** is in direct contact with the outside air surrounding the aircraft **100**. For example, the distal end **152b** of the first line **152** emerges on the upper surface of the wing **104** of the aircraft **100**. In flight, a depression is created at the upper surface of the wing **104**, and, since the first line **152** is fluidically connected by its distal end **152b** to the upper surface of the wing **104**, there is a suction of the outside air surrounding the aircraft **100** into the first line **152**, which maximizes the

air flow rate in said first line **152**.

(42) The first line **152** ensures the evacuation of the air and of the dihydrogen in the case of a leak from the transport line **112a** in the gutter **202** by ensuring an evacuation over the entire length of said gutter **202**. The first part of the first line **152**, between its proximal end **152a** and the bend **154**, also ensures the evacuation of the water from the gutter **202**. Indeed, the bend **154** corresponds to a low point of the first line **152**, and it is at the bend **154** that the water is separated, by gravity, from the air and from the dihydrogen.

(43) The evacuation system **150** comprises a second line **156** which has a proximal end **156a** and a distal end **156b**, in which the second line **156** is fluidically connected to the bend **154** by its proximal end **156a** and emerges outside of the aircraft **100** at an outer wall of the aircraft **100**, here of the consumer device **106**, by its distal end **156b**.

(44) Between the proximal end **156a** of the second line **156** and the distal end **156b** of the second line **156**, the second line **156** has a downward slope to ensure the flow of the water or of any other liquid to the outside and thus ensuring the drainage over the entire length of said gutter **202**.

(45) To improve the suction of the air or of the dihydrogen at the distal end **152b** of the first line **152**, the evacuation system **150** comprises a cap **160** which extends outside of the aircraft **100** around the distal end **152b** by having a closed face oriented towards the front of the aircraft **100** and an open face oriented towards the rear of the aircraft **100**. The gases present in the first line **152** are then sucked by the outside air by virtue of the speed of displacement of the aircraft **100**. The cap **160** is fixed to the outer wall of the aircraft **100**.

(46) To prevent the water present in the second line **156** from freezing, the evacuation system **150** comprises a heating element **162**, for example an electrically powered electrical resistor, which is disposed close to said second line **156**.

(47) Here, the heating element **162** is disposed in an aileron **164** in which the second line **156** is enclosed.

(48) To take advantage of the suction of the outside air, the distal end **156b** of the second line **156** is oriented towards the rear of the aircraft **100**.

(49) While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

1. An aircraft comprising: at least one transport line in which dihydrogen flows, between, upstream a tank and downstream a consumer device, for each transport line, an embedded channel comprising a gutter in which said transport line is fixed and having a first end by which said transport line coming from upstream enters into the gutter and a second end by which said transport line going downstream leaves the gutter, and for each second end, an evacuation system fluidically connected between said second end of the gutter and the outside of the aircraft, the evacuation system comprising: a first line with a proximal end, a distal end and a bend between said proximal end and said distal end, the bend constituting a low point for the first line, and the first line being fluidically connected to said second end of the gutter by said proximal end and emerging outside at a first outer wall of the aircraft by said distal end, and a second line with a proximal end and a

distal end, the second line being fluidically connected to the bend by said proximal end and emerging outside the aircraft at a second outer wall of the aircraft by said distal end, and, between the proximal end of the second line and the distal end of the second line, the second line having a downward slope.

2. The aircraft of claim 1, further comprising: a wing, wherein the distal end of the first line opens at an upper surface of the wing.

3. The aircraft of claim 1, wherein the evacuation system further comprises a cap which extends outside of the aircraft around the distal end of the first line by having a closed face oriented towards a front of the aircraft and an open face oriented towards a rear of the aircraft.

4. The aircraft of claim 1, wherein the evacuation system further comprises a heating element disposed closer to said second line than to said first line.

5. The aircraft of claim 1, wherein the distal end of the second line is oriented towards a rear of the aircraft.
